Data Sheet

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Motivation

This model was created by Domingos Romualdo as part of the Capstone Project for Imperial College's Certificate Program in Machine Learning and Artificial Intelligence. The purpose of the project is to develop a framework to solve the Black-Box Optimization Challenge, in which a relatively small number of data points is given for which the value of a target function is known, and the objective is to identify the global optimum for the function in a limited number of further evaluations. The key decision in each stage is to identify the next point for function evaluation, with the purpose of gaining further knowledge of the function behavior with a view to finding out a global maximum at the end of the experiment.

The approach is based on standard Machine Learning techniques, which can be learned from any good textbook in the subject. The author does not have any outside sources of funding, nor any conflicts of interest to declare.

Composition

The initial dataset for each of the experiments was provided by the host of the Certificate Programme, and as such it is not public. However, a comparable set of data points can be generated for any putative target function, and that can be used to implement and evaluate the approach. In fact, the author followed a similar procedure in order to calibrate the parameters of the model, and how those parameters changed over time. Details and code are available in this repository.

Collection process

The experiments on which the approach was based do not involve explicit data collection on my part.

Preprocessing/cleaning/labelling

No pre-processing was carried out in the dataset for which the model was developed, nor in any dataset used to test or simulate the model.

Uses

The model is intended to identify optimum combinations of input values to various functions in settings where exhaustive grid searches are not practical. The model works best in settings where the functions do not present extreme variability, and in particular where regions which do not seem promising given current data do not include local areas of sudden spikes in function values.

Distribution

The code for the experiment is freely available.

Maintenance

The author plans to provide further updates as improvements and modifications are made to the approach.